Example 2.32 Find the sum of $0.40 + 0.43 + 0.46 + \cdots + 1$.

Solution Here the value of n is not given. But the last term is given. From this, we can find the value of n.

Given, $a = 0.40$ and $l = 1$, we find $d = 0.43 - 0.40 = 0.03$.

$$\text{Therefore, } n = \left(\frac{l-a}{d}\right) + 1$$
$$= \left(\frac{1-0.40}{0.03}\right) + 1 = 21$$

Sum of first n terms of an A.P. $S_n = \frac{n}{2}[a + l]$

, $n = 21$. Therefore, $S_{21} = \frac{21}{2}[0.40 + 1] = 14.7$

So, the sum of 21 terms of the given series is 14.7.

Example 2.33 How many terms of the series $1 + 5 + 9 + \ldots$ must be taken so that their sum is 190?

Solution Here we have to find the value of n, such that $S_n = 190$.

First term $a = 1$, common difference $d = 5 - 1 = 4$.

Sum of first n terms of an A.P.

$$S_n = \frac{n}{2}[2a + (n-1)d] = 190$$
$$\frac{n}{2}[2 \times 1 + (n-1) \times 4] = 190$$
$$n[4n - 2] = 380$$
$$2n^2 - n - 190 = 0$$
$$(n - 10)(2n + 19) = 0$$

But, $n = 10$ as $n = -\frac{19}{2}$ is impossible. Therefore, $n = 10$.

Thinking Corner

The value of n must be positive. Why?

Progress Check

State True or False. Justify it.

1. The nth term of any A.P. is of the form $pn+q$ where p and q are some constants.
2. The sum to n^{th} term of any A.P. is of the form $pn^2+qn + r$ where p, q, r are some constants.

Example 2.34 The 13th term of an A.P. is 3 and the sum of first 13 terms is 234. Find the common difference and the sum of first 21 terms.

Solution Given, the 13th term $= 3$ so, $t_{13} = a + 12d = 3$ ...(1)

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